Data Cleaning and Exploratory Data Analysis Using Pandas

Program

```
# Data Cleaning and Exploratory Data Analysis using Pandas (with user input)
import pandas as pd


def read_number(message):
    """Read a number. Press Enter without typing to mark it as missing."""
    text = input(message).strip()
    if text == "":
        return None
    return float(text)


n = int(input("Enter number of students: "))

names = []
departments = []
attendance = []
marks = []

for i in range(1, n + 1):
    print(f"\nStudent {i}")
    names.append(input("Enter Name: "))
    departments.append(input("Enter Department: "))
    attendance.append(read_number("Enter Attendance (press Enter if missing): "))
    marks.append(read_number("Enter Mark (press Enter if missing): "))

data = {
    "Name": names,
    "Department": departments,
    "Attendance": attendance,
    "Mark": marks
}
df = pd.DataFrame(data)
print("\n--- Original Dataset ---")
print(df)

print("\nMissing Values:")
print(df.isnull().sum())

# Missing value treatment
df["Attendance"] = df["Attendance"].fillna(df["Attendance"].mean())
df["Mark"] = df["Mark"].fillna(df["Mark"].mean())

# Remove duplicate records
df = df.drop_duplicates()
print("\n--- Cleaned Dataset ---")
print(df)

print("\n--- Descriptive Statistics ---")
print(df.describe())

print("\n--- Department-wise Average Marks ---")
print(df.groupby("Department")["Mark"].mean().round(2))

print("\n--- EDA Summary ---")
print("Highest Mark :", df["Mark"].max())
print("Lowest Mark :", df["Mark"].min())
print("Average Mark :", round(df["Mark"].mean(), 2))
print("Average Attendance :", round(df["Attendance"].mean(), 2))
```

Output

```
Enter number of students: 5

Student 1
Enter Name: Asha
Enter Department: CSE
Enter Attendance (press Enter if missing): 90
Enter Mark (press Enter if missing): 88

Student 2
Enter Name: Bala
Enter Department: CSE
Enter Attendance (press Enter if missing): 80
Enter Mark (press Enter if missing): 76

Student 3
Enter Name: Charan
Enter Department: ECE
Enter Attendance (press Enter if missing):
Enter Mark (press Enter if missing): 92

Student 4
Enter Name: Divya
Enter Department: ECE
Enter Attendance (press Enter if missing): 95
Enter Mark (press Enter if missing):

Student 5
Enter Name: Eshan
Enter Department: CSE
Enter Attendance (press Enter if missing): 85
Enter Mark (press Enter if missing): 84

--- Original Dataset ---
     Name Department  Attendance  Mark
0    Asha        CSE        90.0  88.0
1    Bala        CSE        80.0  76.0
2  Charan        ECE         NaN  92.0
3   Divya        ECE        95.0   NaN
4   Eshan        CSE        85.0  84.0

Missing Values:
Name          0
Department    0
Attendance    1
Mark          1
dtype: int64

--- Cleaned Dataset ---
     Name Department  Attendance  Mark
0    Asha        CSE        90.0  88.0
1    Bala        CSE        80.0  76.0
2  Charan        ECE        87.5  92.0
3   Divya        ECE        95.0  85.0
4   Eshan        CSE        85.0  84.0

--- Descriptive Statistics ---
       Attendance      Mark
count     5.00000   5.00000
mean     87.50000  85.00000
std       5.59017   5.91608
```

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min      80.00000  76.00000
25%      85.00000  84.00000
50%      87.50000  85.00000
75%      90.00000  88.00000
max      95.00000  92.00000

--- Department-wise Average Marks ---
Department
CSE    82.67
ECE    88.50
Name: Mark, dtype: float64

--- EDA Summary ---
Highest Mark : 92.0
Lowest Mark : 76.0
Average Mark : 85.0
Average Attendance : 87.5
```